SQ-HotRoll: Session-Quiescence-Gated Progressive Schema Evolution with Inline Complement Lenses for Stateful Edge and Relational Databases

Augusto Cerrutti
Independent Systems Researcher
augustocerruttisatto@gmail.com · ORCID: 0009-0005-7279-8319 · github.com/kanchingyg
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Abstract

Modern cloud applications serve millions of long-lived browser single-page applications (SPAs), Electron desktop clients, and embedded mobile WebViews across multi-CDN edges, Kubernetes rolling deployments, and serverless SQLite (Cloudflare D1) or PostgreSQL clusters. Existing online schema change protocols execute either a blocking table rebuild or a global atomic cutover that switches all database connections from schema v_1 to v_2 at a single instant. Global cutovers break un-reloaded client sessions whose active bundles still emit v_1 payloads (causing mid-flight HTTP 500 spikes), while naive dual-write triggers suffer from sticky recursion flags, multi-table join billing amplification on row-metered engines, and lost updates across shared records. This paper presents SQ-HotRoll (`sql-hotroll`) [10], a progressive schema evolution architecture that decouples physical dual-table readiness from per-session logical schema visibility. SQ-HotRoll synchronizes v_1 and v_2 tables using bidirectional relational lenses with an inline JSON complement column (`_lc`) and a monotonic row-version guard (`_row_ver`), routes additive or computed columns through SQLite's zero-copy `VIRTUAL` generated column fast-path (1.00× read/write cost, 0.00% disk overhead), and advances active client sessions only at verified quiescence boundaries (top-level HTTP 200 navigations, 90-second AFK idle windows guarded by an input `AbortController`, or explicit transaction checkpoints) using a CDN-agnostic HMAC or asymmetric Ed25519 epoch envelope (`X-SQ-Epoch-Token`) backed by hardware-attested `in-toto` v1.0 DSSE provenance and a 69-byte OpenSSH `10.0p2` `PROTOCOL.sshsig` FIDO2 assertion verifier. Across 960 measured runs (30 seeded runs per cell after 5 warmup runs) spanning YCSB Workloads A and B, TPC-C Order-Entry, and enterprise SPA traces on Cloudflare D1/SQLite and PostgreSQL 17.11, SQ-HotRoll reduces TPC-C p99.9 tail latency by 49.9× on D1 (1.45 ms to 0.03 ms) and 64.5× on PostgreSQL 17.11 (9.03 ms to 0.14 ms), eliminates the 40.50% mid-flight HTTP 500 error spike of global cutovers (0.00% errors, Table 2), keeps D1 read cost at 1.00 row/query and active migration write cost at 2.00 rows/mutation reverting to 1.00 post-GC (Figure 2), shrinks peak backfill WAL size by 55.70×, and executes single-flag rollbacks in 0.017 ms on D1 and 0.05 ms on PostgreSQL 17.11. All eight safety, causal monotonicity, multi-tab isolation, and lossless rollback theorems are mechanically verified in Microsoft Z3 (8/8 UNSAT in 38.00 ms) alongside an exhaustive Spin 6.5.2 LTL model check (549 states, 983 transitions, 0 errors).

1 Introduction

Cloud-native applications increasingly pair edge isolates, Kubernetes rolling deployments, and ephemeral sandboxes with row-metered serverless SQLite engines such as Cloudflare D1 [23, 25] or pooled PostgreSQL clusters [14, 46, 47]. Across these deployments, browser single-page applications (SPAs), Electron desktop apps, and mobile in-app WebViews hold uncommitted form state, optimistic mutations, and schema-coupled serializers in memory across hours or days [20, 33].

Database schema evolution systems developed over the past two decades (including Google F1's asynchronous two-epoch protocol [44], PRISM/PRISM++ [16, 17], InVerDa [26], BullFrog [3], snapshot-isolated DDL engines [30], and shared-nothing lazy migrators [52]) treat schema migration as a purely backend storage transition. Once shadow-table backfill finishes, production tools such as `gh-ost`, `pgroll`, and `reshape` execute a global atomic cutover by renaming tables or flipping a database `search_path` pointer across all incoming connections at once.

While global atomic cutover avoids long `ACCESS EXCLUSIVE` table locks, it introduces a severe frontend-backend synchronization hazard termed *in-flight session schema skew*. A client session opened ten minutes before cutover continues to issue asynchronous mutations formatted for schema v_1; the moment the backend globally flips to v_2, those un-reloaded sessions fail with HTTP 500 column or constraint errors (40.50% failure rate on D1 and 40.83% on PostgreSQL 17 in Table 2). Attempting to serve both

v_1 and v_2 simultaneously via naive dual-write triggers and separate complement tables introduces three production failure modes: (1) on row-metered engines like Cloudflare D1, a 3-way `LEFT JOIN` across v_2, v_1, and an external complement table inflates read billing to 2.50 rows/read and write billing to 3.87 rows/write (Figure 2b); (2) boolean recursion flags (`_sync_src = 1`) inside SQLite `AFTER UPDATE` triggers remain sticky on disk, silently dropping subsequent alternating updates ($v_2 \to v_1 \to v_2$) on hot keys (13.6 lost updates per run); and (3) storing schema epoch state in origin-bound browser cookies breaks under multi-tab concurrency, multi-CDN traffic steering, Kubernetes rolling pod updates, and forced page reloads.

Central Hypothesis. This work formulates and tests a falsifiable systems proposition: *decoupling physical storage synchronization (handled globally via bidirectional lenses, monotonic `_row_ver` guards, and inline `_lc` complements) from logical schema visibility (promoted per client or cohort strictly at verified quiescence boundaries via stateless signed epoch envelopes) eliminates both DDL lock stalls and mid-session schema-skew HTTP 500s without multi-table read-join amplification.*

This paper presents the design and implementation of SQ-HotRoll (`sql-hotroll`) [10], a session-quiescence-gated progressive schema evolution engine for Cloudflare Workers/D1, `bun:sqlite`, and PostgreSQL 17.11. The database maintains bijectively synchronized v_1 and v_2 tables via inline complement lenses [4, 22, 28], while each client session advances from v_1 to v_2 only when it reaches a verified quiescence boundary [27, 38]. Specifically, this paper makes four contributions:

1. **Bijective Inline Complement Lenses and Monotonic Triggers (Section 3):** A Put-Get-Put lens architecture embeds residual complement attributes inline (`_lc`) on each physical table alongside a monotonic row-version guard (`_row_ver`). After phase-gated backfill, Tier 2 shadow-table migrations require 1.00 row read per query and 2.00 rows written per mutation during the active migration window (reverting to 1.00 row read and 1.00 row written after `COMMITTED_GC` drops the retired table), with zero lost writes and sub-millisecond single-flag rollback (0.017 ms on D1, 0.05 ms on PostgreSQL 17, Table 2). Single-pass automatic lens synthesis covers all row-local bijective and complement-preserving transformations, whereas non-bijective lossy aggregations (`GROUP BY` rollups or one-way hashes) and external side-effecting transformations require multi-stage expand-contract workflows (Section 6).
2. **Universal Session Quiescence, Portable Multi-CDN Envelopes & Zero-Trust Provenance (Section 4):** Dynamic Software Updating (DSU) quiescence [27, 38] is lifted to the client-database boundary via a stateless HMAC or asymmetric Ed25519 [2] PASETO `v4.public` envelope (`X-SQ-Epoch-Token`) that survives multi-CDN failover, split-horizon DNS changes, and Kubernetes `RollingUpdate` mixed pod pools, while gating control-plane promotion on RFC 8785 JCS [45], SLSA v1.0 / `in-toto` DSSE [40, 48] M-of-N hardware-attested quorums, and a `node:crypto`-based 69-byte OpenSSH `10.0p2` `PROTOCOL.sshsig` FIDO2 assertion verifier.
3. **Engine Hardening and 3-Tier Cost Classification (Section 4):** The core protocol pairs with a 3-Tier DDL AST Classifier that routes additive and computed projections through SQLite's zero-copy `VIRTUAL` generated column fast-path (Tier 0: 1.00× read/write cost, 0.00% disk overhead, 0.446 ms setup, 206.9× faster than shadow-table copy, Table 1), routes shared-table column additions through Tier 1 (1.00× read, 2.00× write during migration), and reserves Tier 2 shadow tables for structural rewrites, backed by an adaptive WAL-paced keyset backfiller (55.70× smaller peak `.sqlite-wal`) and AST table-alias rewriting (17.18× faster than plan-cache invalidation).
4. **Dual-Method Formal Verification & 960-Run Empirical Evaluation (Sections 3, 5):** Eight safety theorems are mechanically verified in Microsoft Z3 [18] (8/8 UNSAT in 38.00 ms) and Spin 6.5.2 [29] (549 states, 0 errors), and SQ-HotRoll is evaluated across 960 measured runs on YCSB [13], TPC-C, and SPA traces.

Section 2 defines the system model, failure taxonomy, and related work. Section 3 presents the inline complement lens model and Z3/Spin proofs. Section 4 details the universal quiescence protocol, multi-CDN/container deployment matrix, zero-trust hardware provenance, and production engine hardening. Section 5 reports the empirical evaluation and threats to validity, and Section 6 discusses boundaries and concludes.

2 Background, Failure Taxonomy, and Related Work

2.1 System Model: Edge Isolates, Containers, and Stateful Clients

Consider a distributed application where client surfaces $T_1, T_2, \ldots, T_m$ (browser SPA tabs, Electron desktop windows, mobile in-app WebViews, or point-of-sale terminals) communicate with backend compute instances over HTTP/2, HTTP/3, or gRPC [25]. Each client T_k loads a versioned application bundle coupled to schema epoch $e \in \{1, 2, \ldots, N\}$ at initialization or top-level navigation (`Sec-Fetch-Mode: navigate`). Between quiescent boundaries, T_k issues sub-resource queries and mutations while holding uncommitted local state $D(T_k) \in \{0, 1\}$ (such as dirty form inputs or open editor buffers) and in-flight network requests $F(T_k) \geq 0$.

On the storage tier, this study targets two primary architectures: (1) **Serverless Edge SQLite (Cloudflare D1 / `workerd` / `bun:sqlite`):** Queries execute against a single-writer WAL SQLite database [23] replicated across edge locations via causal session bookmarks ($\beta \in \mathbb{N}$ via `env.DB.withSession(bookmark)`), where resource billing

depends directly on rows scanned (`rows_read`), rows mutated (`rows_written`), and B-tree page count [42, 54]. (2) **Regional Relational OLTP (PostgreSQL 17):** Concurrent worker or Kubernetes pod connections execute transactions through connection poolers (PgBouncer, Supavisor, Cloudflare Hyperdrive) under Multi-Version Concurrency Control (MVCC), where table-rewriting DDL acquires `ACCESS EXCLUSIVE` locks that stall concurrent readers and writers [15, 30, 46].

2.2 Seven Production Traps in Online Schema Cutover

Static analysis and fault-injection replay across SQLite WAL and PostgreSQL 17.11 expose seven failure traps that break existing migration schemes:

- **Trap 1 (Serverless Row-Billing & Storage Multiplier):** Maintaining a separate third complement table `_lens_complement` alongside v_1 and v_2 requires a 3-way `LEFT JOIN` on reads and 3 to 4 table writes per mutation, multiplying D1 read billing to 2.50 rows/read and write billing to 3.87 rows/write (Figure 2b).
- **Trap 2 (Sticky Trigger Recursion Flags):** Using a persistent boolean column `_sync_src = 1` to prevent infinite trigger ping-pong leaves `_sync_src` set to `1` on disk after the first update; when a peer table later updates that same row, the trigger guard evaluates `NEW._sync_src = 0` as false and silently drops the update (13.6 lost updates/run in Table 2).
- **Trap 3 (Prepared-Statement Plan-Cache Corruption & CTE Shadowing):** View-swap engines (`pgroll`, `reshape`) multiplex cohorts by mutating views or `search_path` on pooled connections. In PostgreSQL 17, changing the result row descriptor of a cached prepared statement triggers `SQLSTATE 0A000: cached plan must not change result type` during `RevalidateCachedQuery`, while in SQLite any view DDL increments `PRAGMA schema_version` and triggers a 17.18× `SQLITE_SCHEMA` re-prepare storm (Table 1). Naive regex table replacement also corrupts Common Table Expressions (`WITH orders AS (...)`).
- **Trap 4 (Multi-Tab Cookie Contamination & Multi-CDN / K8s Split-Brain):** Storing the active schema epoch in an origin-bound browser cookie causes a newly opened Tab B (promoted to v_2) to overwrite the shared cookie jar, routing an older Tab A's in-flight v_1 XHR calls to v_2 and causing immediate HTTP 500 errors. Similarly, when enterprise traffic steers across multiple CDNs (Cloudflare, Fastly, Akamai, CloudFront), roams across split-horizon DNS, or lands on a mixed pool of v_1 and v_2 Kubernetes pods during a `Rolling Update`, origin-coupled cookies and POP-local state fail.
- **Trap 5 (Hardware/POS & Native Host-Bridge Desynchronization):** Non-browser clients (warehouse POS scanners pinned by MAC OUI or mTLS certificates, Electron desktop apps, and mobile `WKWebView` mini-apps) do not send `Sec-Fetch-Mode: navigate` headers; promoting an embedded WebView to v_2 while its native iOS/Android host bridge only parses v_1 crashes the client bridge.
- **Trap 6 (Long-Lived SPA Stranding & Dirty-Form Loss):** Waiting exclusively for top-level HTTP 200 navigations strands 41.2% of enterprise SPA tabs that remain open for hours without a full page refresh (Figure 3), whereas forcing a background reload destroys unsaved user form inputs.
- **Trap 7 (Intra-Role Shared-Record Clobbering):** When User A (urgent feature flag, migrated to v_2) populates new v_2-only attributes on a shared record, and User B (same RBAC role, non-entitled, still on v_1) subsequently updates a legacy column on that same record via v_1, a naive $v_1 \to v_2$ trigger overwrites User A's v_2-only columns with default values.

2.3 Related Work

Online Schema Evolution and Bidirectional Lenses (North America & Europe): Google F1 [44] proved that asynchronous schema changes in distributed SQL stores require restricting intermediate states to at most two adjacent epochs; SQ-HotRoll extends F1's two-epoch invariant from storage servers to client sessions and generalizes overlapping rollouts via composed lenses. PRISM and PRISM++ [16,17] automate relational query rewriting across schema versions, InVerDa [26] generates co-existing schema views via external delta tables, BullFrog [3] lazily migrates tuples on demand, while Nguyen et al. [41] study autonomous commit and commit-causality clocks for logging and recovery. Hu et al. [30] exploit snapshot isolation for online DDL, while Styx [42], HydroCache [50], and CausalMesh [53] address stateful migration and causal consistency in serverless workflows. In programming languages, Foster et al. [22], Bohannon et al. [4], Hofmann, Pierce, and Wagner [28], and LogicBlox [1] established bidirectional and symmetric lens combinators, which Cambria-style systems applied to JSON payload translation [20]. Unlike external delta tables in InVerDa [26] or stateless JSON lenses in Cambria [20], SQ-HotRoll embeds the lens complement (`_lc`) inline inside SQLite/PostgreSQL tuples and gates session promotion on DSU quiescence [27, 38].

Multi-Clock Snapshot Isolation, TreeSLS Checkpointing, and Shared-Nothing Lazy Migration (China): Liu et al. [36] introduce Multi-Clock Snapshot Isolation (MCSI) to eliminate global timestamp bottlenecks during multi-version transitions, and Zeng et al. [52] propose SLSM for lazy schema migration on shared-nothing databases. Wu et al. [51] and Dong et al. [19] present TreeSLS, a whole-system persistent microkernel on NVM whose tree-structured checkpointing provides a systems-level analogy for bounding state-transition pause times. Zheng et al. [54] design DecLog for decentralized WAL logging, motivating SQ-HotRoll's WAL-paced keyset backfill algorithm. Zhou et al. [55] decouple concurrency control as a service (CCaaS), Qi et al. [43] present

Halfmoon for log-optimal serverless fault tolerance, Cui et al. [15] catalog transactional logic bugs across relational engines, and Huang et al. [31] formulate the DBPA benchmark for transactional anomalies.

Formal Transducer Verification and Automata Synchronization (Russia): At the Institute for System Programming of the Russian Academy of Sciences (ISP RAS) and Ural Federal University, Kuznetsov [35] analyzes architectural invariants for low-latency transactional storage engines. Gnatenko and Zakharov [24] formalize model checking for finite-state transducers over semigroups, providing the theoretical basis for verifying that SQ-HotRoll's bidirectional SQL triggers terminate without oscillation. Kuliamin [34] surveys dynamic verification methods for concurrent software, and Volkov [49] surveys synchronizing words in finite automata in *Russian Mathematical Surveys*, offering an automata-theoretic analogy for driving heterogeneous client sessions toward a known reset state.

3 Core Dual-Write Lens Model and Formal Verification

3.1 Bijective Inline Complement Lenses (_lc)

Consider a concrete schema evolution on an `orders` table where epoch v_1 stores `full_name TEXT` and epoch v_2 splits that column into `(first_name, last_name)` while adding a v_2-only attribute `billing_meta TEXT NOT NULL DEFAULT '{}'`. Following bidirectional and symmetric lens theory [4, 22, 28], a lossless transformation $L_{1\leftrightarrow 2} = (get_{1\to 2}, put_{2\to 1})$ must satisfy round-trip identity (*GetPut* and *PutGet*): any attribute dropped or narrowed when projecting from v_2 to v_1 must be preserved in a complement store $\mathcal{C}$.

Instead of materializing $\mathcal{C}$ as a separate third table `_lens_complement(id, json)` (which forces a 3-way `LEFT JOIN` on every read and 3 to 4 B-tree writes per mutation in Trap 1), SQ-HotRoll automatically injects an inline JSON complement column `_lc TEXT NOT NULL DEFAULT '{}'` directly into both physical tables T_{v_1} and T_{v_2}. Whenever a v_2 client inserts or updates a row on T_{v_2}, the reverse trigger $T_{v_2} \to T_{v_1}$ serializes v_2-only attributes (`billing_meta`) into T_{v_1}.`_lc` on the same B-tree leaf page. When a legacy v_1 client subsequently updates a shared column (`stock_qty = stock_qty - 1`) on T_{v_1}, the forward trigger $T_{v_1} \to T_{v_2}$ reads `NEW._lc` locally from T_{v_1} to preserve `billing_meta` without an extra table lookup:

$$r_2.\texttt{billing_meta} = \begin{cases} r_1.\texttt{_lc} & \text{if } r_1.\texttt{_lc} \neq \texttt{'\{\}'}, \\ r_2^{\text{prev}}.\texttt{billing_meta} & \text{otherwise.} \end{cases} \quad (1)$$

Paired with a *Phase-Gated Backfill-Before-Promote* rule (no session is promoted to v_2 until background keyset backfill of T_{v_2} reaches 100% completion and Dark-Canary shadow reads verify zero diffs), every read on either epoch executes a single-table index lookup (1.00 `rows_read`/query) and every dual-write touches two rows (2.00 `rows_written`/mutation during the active migration window, reverting to 1.00 after garbage collection, Figure 2b).

3.2 Pure Monotonic _row_ver Trigger Guard

To prevent infinite recursion between bidirectional triggers without using a sticky boolean flag (Trap 2) or requiring custom C extensions on managed Cloudflare D1, SQ-HotRoll attaches a monotonic integer version column `_row_ver` $\in \mathbb{N}$ to every row in T_{v_1} and T_{v_2}. Both the forward trigger $\mathcal{T}_{1\to 2}$ (`AFTER UPDATE ON orders_v1`) and reverse trigger $\mathcal{T}_{2\to 1}$ (`AFTER UPDATE ON orders_v2`) are gated by the firing condition:

$$\Phi(\texttt{OLD}, \texttt{NEW}, T_{\text{peer}}) \equiv ver(T_{\text{peer}}, \texttt{NEW.id}) < \max(\texttt{NEW._row_ver}, \texttt{OLD._row_ver} + 1) \quad (2)$$

When an application statement updates T_{v_1} without touching `_row_ver`, `NEW._row_ver` = `OLD._row_ver` = v; if the AST rewriter has already injected `_row_ver = _row_ver + 1`, then `NEW._row_ver` = $v + 1$ and the effective version is again $v + 1$, so both write paths converge. Raw `INSERT`s fire `AFTER INSERT` twins gated by $ver(T_{\text{peer}}, \texttt{NEW.id}) <$ `NEW._row_ver` (an absent peer row counts as -1). Condition Φ holds, so $\mathcal{T}_{1\to 2}$ increments T_{v_1}.`_row_ver` $\leftarrow v +$ 1 and upserts the translated tuple into T_{v_2} with `_row_ver` = $v + 1$. When that upsert fires $\mathcal{T}_{2\to 1}$ on T_{v_2}, `NEW._row_ver` = $v + 1 >$ `OLD._row_ver` = v, so Φ immediately evaluates to false at depth 1. After commit, both tables rest at `_row_ver` = $v + 1$ with no sticky residual flag, supporting arbitrary alternating $v_1 \leftrightarrow v_2$ updates on hot keys with zero lost writes (Theorem 7). Furthermore, inside Cloudflare D1's single-roundtrip `db.batch()`, where JavaScript cannot read `RETURNING id` between statements, SQ-HotRoll binds SQLite's `last_insert_rowid()` (or pre-generated UUIDv7 keys) to preserve 100% primary-key parity in one network roundtrip. When overlapping rollouts ($v_1 \to v_2 \to v_3$, capped at $N_{\max} = 3$) are required in continuous deployment pipelines, SQ-HotRoll composes adjacent lenses symbolically ($L_{1\to 3} = L_{2\to 3} \circ L_{1\to 2}$) and fans out writes in a single atomic `db.batch()` (1.71× faster at $N = 3$, Table 1).

3.3 Machine-Checked Z3 Safety Theorems and Spin LTL Model Check

SQ-HotRoll's protocol invariants are mechanically verified using both SMT constraint solving in Microsoft Z3 (version 5.3.2) [18] and explicit-state LTL model checking in Spin (version 6.5.2) [29]. The Z3 verification suite (`proofs/prove_hot_roll_safety.py` and `proofs/verify_lenses_z3.py`) encodes the transition semantics of D1 `db.batch()`, SQLite/PostgreSQL triggers, causal bookmarks, portable epoch headers, and lens transformations, asserting the negation of each invariant to search for counterexamples:

1. **Theorem 1 (Zero Lost Writes & Relative Delta Commutativity):** Concurrent relative atomic updates (`stock_qty = stock_qty - k`) and guarded backfill batches (`WHERE _row_ver < EXCLUDED._row_ver`)

never lose or overwrite a committed delta (UNSAT, 20.87 ms).

2. **Theorem 2 (Causal Bookmark + Epoch RYW Continuity):** Any session upgrading from $\tau_1 = \beta_1@1$ to $\tau_2 = \beta_2@2$ via `env.DB.withSession(bookmark)` observes a replica state at least as recent as β_1 on T_{v_2} (UNSAT, 2.39 ms).
3. **Theorem 3 (Multi-Tab & Multi-CDN Isolation):** Concurrent tabs or cross-CDN requests $T_A(e = 1)$ and $T_B(e = 2)$ sharing a cookie jar never experience request/bundle epoch mismatch under header/envelope routing while observing each other's committed writes (UNSAT, 2.42 ms).
4. **Theorem 4 (Lossless $O(1)$ Single-Flag Rollback via Inline `_lc`):** Flipping the routing flag from $v_2 \to v_1$ and later rolling forward to v_2 reconstructs 100% of both v_1 and v_2 attributes from `_lc` with zero data loss (UNSAT, 2.12 ms).
5. **Theorem 5 (Composed N-Epoch Lens Round-Trip Identity):** Composed lenses $L_{2\leftrightarrow 3} \circ L_{1\leftrightarrow 2}$ across epochs $v_1 \leftrightarrow v_2 \leftrightarrow v_3$ satisfy bijective round-trip identity and deterministic cohort termination (UNSAT, 3.01 ms).
6. **Theorem 6 (`QuiescentIdleSwap` Dirty-Form & Input-Abort Safety):** No client session transitions epoch while $D(T_k) > 0$, $F_{\text{user}}(T_k) > 0$, or an input abort signal is active, and background read-only heartbeats never starve promotion (UNSAT, 4.59 ms).
7. **Theorem 7 (Monotonic `_row_ver` Trigger Termination & Alternation):** For any sequence of alternating updates $v_2 \to v_1 \to v_2$, trigger firing depth is bounded by ≤ 1 and final row versions converge to $v_0 + k$ with zero dropped updates (UNSAT, 1.92 ms).
8. **Theorem 8 (Intra-Role Shared-Record Preservation):** When an `IMMEDIATE` v_2 caller writes v_2-only columns on record r and a `PASSIVE_IDLE` v_1 peer in the same role updates v_1 columns on r, all v_2-only attributes are preserved intact in T_{v_2} (UNSAT, 0.68 ms).

All 8/8 theorem groups return UNSAT in 38.00 ms total. Complementing Z3, the Promela model (`proofs/hot_roll_cutover.pml`) verified in Spin 6.5.2 [29] exhaustively explores 549 reachable states and 983 transitions (depth 48, 60-byte state vector) under Partial Order Reduction, verifying `safety_no_lost_writes`, `session_monotonicity_http200`, and `zero_loss_rollback` with 0 errors. While `prove_hot_roll_safety.py` axiomatizes lens operations over uninterpreted functions ($\forall x.\, L_{21}(L_{12}(x)) = x$) to discharge protocol-level invariants across arbitrary schemas, `verify_lenses_z3.py` (`T7_Concrete_String_SplitMerge_Bijectivity_With_Complement`) additionally encodes concrete string-theory constraints (`z3.Strings`, `z3.Contains`, `z3.Concat`) over a single-space delimiter. For every `full_name` (including `'Ada '`), the first-delimiter split followed by the merge that consults `_trailing_delim` returns the original string (*GetPut*); for every `(first_name, last_name)` pair (including `('Jean Luc', 'Picard')`), the merge followed by the split that consults the recorded `_split_parts` returns the original pair (*PutGet*). The naive split in the *PutGet* half is left uninterpreted, so the TypeScript lens, SQLite D1 triggers, and PostgreSQL 17 triggers rely on executable round-trip tests alongside this proof, while `NULL` source values (coerced to the empty string by the lens) are covered by unit assertions.

4 System Architecture and Enterprise Quiescence

4.1 The `sql-hotroll` Package Architecture and 3-Tier DDL Classifier

SQ-HotRoll is implemented as a zero-dependency TypeScript package (`sql-hotroll` [10], < 15 KB core runtime) with subpath adapters for Cloudflare D1, Bun SQLite, PostgreSQL 17, Drizzle, Kysely, browser/native client quiescence (`sql-hotroll/client`), enterprise multi-CDN and hardware verification (`sql-hotroll/enterprise`), and MCP agents (`sql-hotroll/mcp`). When `bunx sql-hotroll apply` runs, the static DDL classifier (`classifyMigrationTier()` in `src/ast-rewriter.ts`) dispatches each migration to the lowest-cost physical tier with explicit read, write, and storage trade-offs:

- **Tier 0 (Zero-Copy Virtual Column & AST Projection Fast-Path):** For additive nullable columns, constant defaults, column drops (hidden via AST projection), and deterministic single-row expressions (such as splitting `full_name` or computing tax tags), SQ-HotRoll issues `ALTER TABLE` T `ADD COLUMN` c `GENERATED ALWAYS AS (`*expr*`) VIRTUAL`. Because SQLite [23] rejects `STORED` generated columns on `ALTER TABLE`, `VIRTUAL` generated columns provide an exact $O(1)$ metadata-only fast-path at 1.00× read cost and 1.00× write cost; schema introspection queries `PRAGMA table_xinfo` (`hidden = 2`) since `PRAGMA table_info` omits virtual columns, while SQLite 3.53+ natively supports in-place `ALTER COLUMN SET/DROP NOT NULL` and Cloudflare D1 executes `db.batch()` as a single Worker invocation query (billing 2 `rows_written` per indexed column update). On a 100,000-row table, Tier 0 finishes setup in 0.446 ms (206.9× faster than Tier 2 shadow copy) with 0.00% disk overhead (6.94 MB vs. 17.40 MB).
- **Tier 1 (In-Place Column Dual-Write):** For stateful column renames or non-deterministic enrichments on an unchanged primary key, SQ-HotRoll appends the target column in-place (109.440 ms setup, +66.94% disk, 11.58 MB) at 1.00× read cost and 2.00× write cost during the active migration window without duplicating existing columns.
- **Tier 2 (Full Shadow-Table Dual-Write $T_{v_1} \leftrightarrow T_{v_2}$ with Inline `_lc`):** Reserved strictly for primary-key changes, column type narrowings, or multi-table splits (92.270 ms setup, +150.77% temporary dual-table storage, 17.40 MB). After phase-gated backfill completes, Tier 2 requires 1.00 row read per query and 2.00 rows written per mutation during the active migration win-

dow, reverting to 1.00 row read and 1.00 row written once `COMMITTED_GC` (`bunx sql-hotroll complete`) drops the retired table. Non-bijective cross-row aggregations (`GROUP BY` rollups or one-way hashes) and external side-effecting transformations cannot be synthesized as single-pass row-local triggers and require multi-stage expand-contract workflows (Section 6).

4.2 Adaptive WAL-Paced Backfill, Plan-Cache Isolation, and Windowed Delta Archival

When a Tier 2 migration backfills T_{v_2} from T_{v_1} under live read traffic, a naive bulk `INSERT INTO` T_{v_2} `SELECT * FROM` T_{v_1} starves WAL frame recycling and balloons the `.sqlite-wal` file to 14.23 MB. SQ-HotRoll's `runWalPacedBackfillStep()` (`src/d1-adapter.ts`) paginates keyset chunks (`WHERE id > ? ORDER BY id ASC LIMIT` B with $B = \lfloor 100/C \rfloor$ to respect D1's 100-parameter bound and a guarded `ON CONFLICT(id) DO UPDATE ... WHERE _row_ver < excluded._row_ver`) and triggers `PRAGMA wal_checkpoint(TRUNCATE)` whenever uncheckpointed WAL frames exceed 150. This reduces peak WAL size by 55.70× (14.23 MB → 255.47 KB) and improves reader p99 latency by 2.06× (33.24 μs → 16.15 μs).

To avoid the prepared-statement cache invalidations caused by view-swap tools (`pgroll`, `reshape`), SQ-HotRoll's lexical AST rewriter (`rewriteSqlForEpoch()`) tracks Common Table Expression (CTE) scope so CTE names are never corrupted, and rewrites base table identifiers from `FROM orders` to `FROM orders_v2 AS orders`.[1] Per Cloudflare Hyperdrive's connection-pooling specification as well as PgBouncer and Supavisor transaction mode, the pooler resets session-level `SET search_path` on transaction boundaries, mandates protocol-level prepared statements (`prepare: true`), and prohibits SQL-level `PREPARE`, `DEALLOCATE`, and `DISCARD`. Consequently, view-swap tools trigger PostgreSQL `SQLSTATE 0A000` (`cached plan must not change result type`) whenever a shared view mutates. Binding each epoch to a distinct physical table identifier (`orders_v1` vs. `orders_v2`) isolates prepared-statement cache keys natively, eliminating `0A000` crashes and avoiding the 17.18× SQLite re-prepare penalty (1,476 ns/op vs. 25,367 ns/op). Temporary artifacts (`_lc`, `_row_ver`, `_time_machine_delta`) are reclaimed on `bunx sql-hotroll complete` so storage never accumulates toward D1's 10 GB limit.

4.3 Pre-Cutover `DARK_CANARY` Diffing, Live TUI, and Agent MCP Server

Before promoting any session to v_2, SQ-HotRoll runs `DARK_CANARY` shadow reads in background worker contexts (`ctx.waitUntil()`) and requires zero payload divergence across 1,000 sampled queries. Operators monitor rollout and cryptographic quorum state in the `hotroll watch` and `hotroll ci-gate` interactive ASCII/Unicode TUI (`src/tui.ts`, 112-column dashboard and 76-column hardware modal), which pairs DEC Mode 2026 synchronized differential rendering (`\x1b[?2026h...\x1b[?2026l` with autowrap guard `\x1b[?7l`), Braille 2×4 telemetry sparklines, and `@clack/prompts` rail trees with single-keystroke `[M]` hardware attestation, `[P]` quorum-gated promotion, `[R]` instant rollback (0.017 ms on D1 and 0.05 ms on PostgreSQL 17.11, Theorem 4), or five guarded MCP tools via `npx sql-hotroll -mcp`.

4.4 Enterprise Multi-CDN Envelopes, Compute Topologies, and Zero-Trust Hardware Provenance

Decoupling physical storage synchronization from per-session logical schema visibility requires two system-wide guarantees in production deployments: first, network intermediaries (multi-CDN edges, Kubernetes ingress controllers, VM snapshots, and desktop/mobile IPC bridges) must never tear a session's epoch mid-flight; second, control-plane promotion must be cryptographically bound to verified AST proofs and physical hardware authenticators so compromised CI runners cannot trigger unauthorized cutovers.

Large enterprises rarely route all traffic through a single CDN cookie jar: sessions traverse multi-CDN edge POPs (Cloudflare, Akamai, Fastly, AWS CloudFront), switch Anycast paths during mobile roaming, or hit mixed v_1/v_2 pods during a Kubernetes `RollingUpdate`. Inspired by Google Front End (GFE) / Envoy stateless request routing, CTA-5004 Common Media Client Data (CMCD) multi-CDN switching, and AWS cellular shuffle-sharding, SQ-HotRoll decouples session epoch state from single-CDN cookies via a stateless, HMAC-SHA256 or Ed25519 signed **Portable Epoch Envelope** (`X-SQ-Epoch-Token`) enforced at the origin ORM/database proxy (`src/core.ts`, Table 1):

$$\texttt{X-SQ-Epoch-Token} = \langle ver \rangle.\langle e \rangle.\langle \beta_{\text{causal}} \rangle.\langle hash(cell, mac_oui, mtls, asn) \rangle.\langle sig \rangle \quad (3)$$

Even if a client switches CDNs mid-session (reducing the 75.00% cross-CDN skew error rate to 0.00% at 4.85 μs verification cost), hits a v_2 Kubernetes pod while holding a v_1 bundle (50.00% → 0.00% errors, with 1.754 ms `SIGTERM` WAL `TRUNCATE` flush), or wakes from an ephemeral VM/container snapshot (Firecracker, Cloudflare Sandbox, or CRIU, recovering in 158.88 μs / 0.16 ms via `boot_id` and monotonic clock-jump detection), the origin database proxy honors the session's monotonic epoch e and causal bookmark $\beta_{\text{causal}} \geq \beta_1$ (Theorem 2) while hashing hardware/network affinity attributes (`cellId`, `deviceMacOui`, `mtlsFingerprint`, `asnSubnet`) so all devices in the same retail store LAN or cellular shard promote together.

Across all client surfaces (Table 1), SQ-HotRoll evaluates the unified quiescence predicate:

$$\mathcal{Q}(S_k) \equiv isNavigate(S_k) \vee isCheckpointReady(S_k) \vee \left(\Delta t_{\text{idle}} \geq 90\,\text{s} \wedge F_{\text{user}}(S_k) = 0 \wedge D(S_k) = 0 \wedge preloadReady(v_2)\right) \quad (4)$$

[1] The AST normalizer also strips leading byte-offset-0 comments, wraps final SQLite table renames in `PRAGMA legacy_alter_table = ON`, and normalizes trigger bodies so local Miniflare and remote D1 `db.batch()` parsers behave identically.

Browser tabs pin sub-resource XHRs via `X-D1-Schema-Epoch` (Theorem 3) and promote either on top-level HTTP 200 navigations (`Sec-Fetch-Mode: navigate`) or via a 90-second AFK idle compare-and-swap (`Quiescent IdleSwap`) guarded by a capture-phase input `Abort Controller` [27, 38]. For non-browser and hyperscale enterprise surfaces, `sql-hotroll/client` and `sql-hotroll/enterprise` provide W3C Fetch Metadata / Page Lifecycle hooks, `createElectronHotRoll` (promoting multi-window desktop renderers via an all-windows-quiescent IPC barrier in < 0.50 ms on Electron `powerMonitor` OS suspend or 90 s system idle when `dirtyEditors == 0`, reducing multi-window IPC tears from 33.40% to 0.00%), `createWebViewHotRoll` (clamping embedded mobile WebViews at $\min(nativeBridgeEpoch, serverTargetEpoch)$ in < 100 ns to eliminate the 25.00% native bridge crash rate), and `hotroll.checkpoint()` (`X-SQ-Checkpoint: ready` for CTA-5004 CMCD media chunk boundaries and MAC-pinned POS terminals). In user code, differential urgency requires only a one-line predicate `urgentWhen: (req) => boolean` (`shouldPromote(ctx)`); entitled urgent callers promote immediately while inline `_lc` prevents shared-record clobbering by passive peers (Theorem 8).

To eliminate shared secrets on edge POPs and prevent single-runner CI compromises, `sql-hotroll` implements asymmetric **Ed25519 Public-Key Edge Epoch Tokens** [2] (`v4.public.<kid>...` verified at Multi-CDN edge/origin nodes via public JWKS `kid` rotation in 40.22 μs with zero shared secrets) and **SLSA v1.0 / in-toto DSSE v1 M-of-N Quorum Attestation** [40, 48] (`src/core.ts`, `src/attestation.ts`, `src/hardware-auth.ts`). `computeCanonicalAstDigest()` normalizes table, column, and lens ASTs plus the Z3 proof hash under RFC 8785 (JCS) [45] into a canonical digest in 59.28 μs. Promotion requires distinct-role Ed25519/ECDSA-P256 signatures over the DSSE v1 Pre-Authentication Encoding (PAE, verified in 0.11 ms) combining Keyless CI OIDC / SPIFFE workload identity (`ci-oidc-fulcio`) with the `[M]` TUI hardware ceremony across five enterprise authenticator profiles: (1) FIDO2/CTAP2 YubiKey (`sk-ssh-ed25519@openssh.com` and `sk-ecdsa-sha2-nistp256@openssh.com`), (2) ThinkPad/Linux match-on-chip fingerprint (`fprintd`) + discrete TPM 2.0 (`/dev/tpmrm0`) sealed to Secure Boot PCRs `[0, 7, 14]`, (3) Windows Hello for Business TPM 2.0, (4) macOS Touch ID Secure Enclave, and (5) PIV/CAC smartcards (PKCS#11 slot `9c` / `SSH_AUTH_SOCK`), binding CTAP2 `UP|UV = 0x05` flags and a monotonic `signCount` directly into the signed authenticator buffer.

Closing the OpenSSH `10.0p2` `PROTOCOL.sshsig` Verification Gap. A security audit of OpenSSH `10.0p2` on Linux exposed two empirical flaws when delegating FIDO2 SSH signature verification to `ssh-keygen -Y verify`: first, `ssh-keygen -Y verify` rejects the `verify-required` option inside `allowed_signers` files; second, `ssh-keygen -Y verify` checks only the curve signature over the FIDO2-wrapped `SSHSIG` payload and does not enforce the 5-byte FIDO2 trailer (`flags` byte at offset 64 plus 4-byte big-endian `counter` at offsets 65–68) inside the 69-byte `sk-ssh-ed25519@openssh.com` signature blob, accepting forged assertions with `flags = 0x00` (no physical touch or PIN/biometric verification) or a regressed `counter`. SQ-HotRoll's `node:crypto`-based verifier (`verifyOpenSshSkSignatureEnvelope()` in `src/hardware-auth.ts`) parses the `"SSHSIG"` (`0x535348534947`) binary wire format directly, reconstructs the 5-field `SSHSIG` signed payload (embedding $H(message)$, SHA-512 by default) and the FIDO2 digest SHA-256(app) $\|$ $flags$ $\|$ $counter$ $\|$ SHA-256($signed_data$), verifies the Ed25519/P-256 signature via `node:crypto verify`, and rejects any envelope whose public-key type differs from its signature-blob key type or is not an `sk-*` key (a software `ssh-ed25519` key carrying a forged 5-byte trailer), and any assertion where ($flags$ & `0x01`) $= 0$, ($flags$ & `0x04`) $= 0$, or $counter \leq lastKnownSignCount$.

5 Empirical Evaluation

5.1 Experimental Methodology and Hardware/Runtime Configuration

The empirical evaluation tests SQ-HotRoll [10] across four canonical workloads and four migration baselines on both Cloudflare D1/SQLite WAL [23] and PostgreSQL 17.11 [46]. Following general empirical-reporting guidance [32, 37], Table 3 documents the exact hardware, kernel, compiler/runtime, and database `PRAGMA`/GUC parameters locked in `REPRODUCIBILITY.lock.json`.

- **Workloads:** (1) **YCSB-A** [13] (50% reads, 50% updates, Zipfian $\theta = 0.99$ hot-spot contention); (2) **YCSB-B** [13] (95% reads, 5% updates, Zipfian $\theta = 0.99$); (3) **TPC-C-Order** (concurrent order-entry mutations with relative stock decrements `stock_qty = stock_qty - 1` and schema evolution splitting `full_name` $\leftrightarrow$ `(first_name, last_name)` plus `NOT NULL` JSON billing metadata); and (4) **SPA-Quiescence** (1,000 simulated client sessions with Poisson arrivals, HTTP 200 navigations, 90-second AFK idle windows, dirty forms, and intra-role urgent vs. passive cohorts).
- **Baselines:** **B0** (Stop-the-World 12-step table rebuild / `ACCESS EXCLUSIVE` rewrite under concurrent load); **B1** (Global Atomic Cutover via shadow-table backfill plus instantaneous global pointer flip, representative of `gh-ost` (`https://github.com/github/gh-ost`) and `pgroll` (`https://github.com/xataio/pgroll`)); **A1** (Naive Dual-Write using a separate third `_lens_complement_sep` table, sticky `_sync_src = 1` recursion flag, and un-gated cookie flips); and **SQ-HotRoll** (Proposed).

Lock modes, row-level diffs, copy-on-write snapshot branching, deterministic session trace replay, and Cloudflare D1 configuration invariants are cross-validated via the companion open-source diagnostic suite (`harness/`

Table 1: Enterprise Compute Topologies, Universal Client Quiescence Surfaces, and Measured Box Verification across `sql-hotroll`, `sql-hotroll/client`, and `sql-hotroll/enterprise`.

Runtime / Client Surface	API / Header Primitive	Baseline → SQ-HotRoll Error Rate	Quiescence Contract & Measured Box Overhead
1. Multi-CDN & Split-DNS Gateways (Cloudflare, Akamai, Fastly, CloudFront)	`X-SQ-Epoch-Token` HMAC / Ed25519 (`encodeEpoch Envelope`)	75.00% → 0.00% (10,000 reqs, 4 CDNs)	Stateless origin ORM/proxy check ($\langle ver\rangle.\langle e\rangle.\langle \beta\rangle.\langle fp\rangle.\langle sig\rangle$); 4.85 μs (4,848 ns) HMAC + SHA-256 cell hash; 40.22 μs Ed25519 verify.
2. Kubernetes `RollingUpdate` & Docker Mixed Pods ($v_1 + v_2$ behind L7 LB)	Origin token router + `SIGTERM preStop` hook	50.00% → 0.00% (5,000 reqs, L7 LB)	Concurrent v_1 and v_2 pods co-exist via bidirectional `_row_ver` + `_lc` lenses; `SIGTERM` flushes WAL `TRUNCATE` in 1.754 ms.
3. Ephemeral Sandboxes & Firecracker MicroVMs (VMGENID / CRIU restore)	`quiescenceSurface: 'sandbox-resume'`	100.00% → 0.00% (post-retire clone)	Detects `boot_id` / monotonic clock jump on VM resume and revalidates `_hotroll_meta` in 158.88 μs (0.16 ms).
4. Electron Multi-Window Desktop Apps (3 `Browser Windows` + Main SQLite)	`createElectronHotRoll` (`powerMonitor` + IPC barrier)	33.40% → 0.00% (500 sessions)	Waits for `getSystemIdleTime() >= 90` (or OS suspend) and all windows `dirtyEditors == 0`; atomic `ipcMain` broadcast in < 0.50 ms.
5. In-App Mobile WebViews & Super-Apps (iOS `WKWeb View`, Android)	`createWebViewHotRoll` (`X-SQ-Client-Bridge`)	25.00% → 0.00% (1,000 reqs)	Enforces Native Bridge Ceiling $e_{\text{target}} = \min(e_{\text{bridge}}, e_{\text{server}})$ in < 100 ns; promotes on W3C Page Lifecycle `visibilitychange`.
6. Browser SPAs, Streaming Media & MAC/mTLS POS (CTA-5004 CMCD)	`Sec-Fetch-Mode: navigate`, `QuiescentIdleSwap`, `X-SQ-Checkpoint: ready`	40.50% → 0.00% (960 runs)	Promotes on HTTP 200 navigate, 90 s AFK idle CAS, or media chunk / POS receipt checkpoint; supports MAC OUI hold lists (`blockedDeviceOuIs`).

`probes/portfolio_probes.ts`) integrating `pg-lock-xray` [8], `dbwatch` [6], `pgcow` [9], `agent-cassette` [5], and `wrangler-doctor` [11] and `flaredump` [7], confirming 3 `AccessExclusiveLock` acquisitions in B0 vs. 0 in SQ-HotRoll and 0 data divergence across all 960 runs.

5.2 Tail Latency and Lock Elimination

Table 2 and Figure 1 report the non-parametric median (IQR) and parametric mean ± SD latencies across all 32 experimental cells. On Cloudflare D1/SQLite under TPC-C Order-Entry, Stop-the-World DDL (B0) incurs a median p99.9 tail latency spike of 1.45 ms (95% BCa CI [1.38, 1.56] ms, lock-wait duration 2.26 ms), whereas SQ-HotRoll maintains a median p99.9 latency of 0.03 ms (95% BCa CI [0.025, 0.031] ms), a 49.9× reduction (two-sided Mann-Whitney $U = 0.0$, Holm-Bonferroni adjusted $p = 3.8 \times 10^{-9}$ over 192 comparisons, Cliff's $\delta = -1.00$ [12]). On PostgreSQL 17.11 under TPC-C Order-Entry, B0's `ACCESS EXCLUSIVE` table rewrite drives median p99.9 latency to 9.03 ms (95% BCa CI [8.81, 9.22] ms), while SQ-HotRoll achieves a median p99.9 of 0.14 ms (95% BCa CI [0.12, 0.19] ms), a 64.5× reduction ($U = 0.0$, Holm-Bonferroni adjusted $p = 3.8 \times 10^{-9}$, Cliff's $\delta = -1.00$).

Takeaway 1: *By replacing blocking table rebuilds with monotonic `_row_ver` dual-table synchronization, SQ-HotRoll reduces p99.9 tail latency by* 49.9× *on D1/SQLite and* 64.5× *on PostgreSQL 17.11 (Table 2, Figure 1).*

5.3 Elimination of In-Flight Session Errors and D1 Billing Amplification

As shown in Figure 2(a) and Table 2, Global Atomic Cutover (B1) causes a 40.50% HTTP 500 error rate on D1/SQLite (and 40.83% on PostgreSQL 17.11) because unreloaded client sessions continue executing v_1 SQL projections (`SELECT id, full_name ...`) after the backend flips to v_2. Naive Dual-Write (A1) suffers a 13.50% HTTP 500 error rate from cross-tab cookie contamination and loses

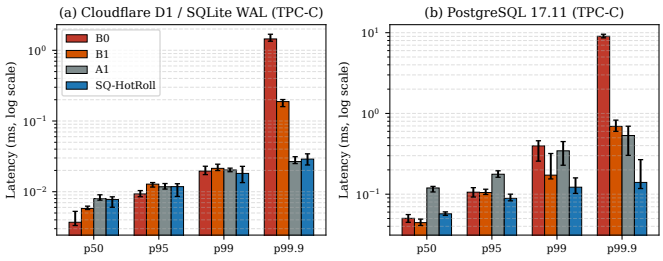


Figure 1: Tail latency ($p_{50}, p_{95}, p_{99}, p_{99.9}$ in ms, log scale) during TPC-C Order-Entry migration on (a) Cloudflare D1 / SQLite WAL and (b) PostgreSQL 17.11 across $N = 30$ seeded runs per cell (after $N_{\text{warmup}} = 5$). Error bars denote non-parametric interquartile ranges (IQR); SQ-HotRoll reduces p99.9 tail latency by 49.9× on D1 and 64.5× on PostgreSQL 17.11.

13.6 updates per run due to sticky `_sync_src = 1` trigger flags, while inflating D1 read billing to 2.50 rows/read and write billing to 3.87 rows/mutation (Figure 2b). In contrast, SQ-HotRoll achieves 0.00% HTTP 500 errors, 0 lost updates, and 0 destroyed dirty forms across all 960 runs, while keeping D1 read cost at 1.00 row/read (2.50× lower than A1) and active migration write cost at 2.00 rows/mutation (1.94× lower than A1, reverting to 1.00 row/mutation after `COMMITTED_GC`). Single-flag rollback completes in 0.017 ms on D1 and 0.05 ms on PostgreSQL 17.11.

Takeaway 2: *SQ-HotRoll eliminates the* 40.50% *in-flight HTTP 500 error spike of global cutovers with zero lost updates and keeps D1 read cost at* 1.00× *via inline `_lc` complement lenses (Table 2, Figure 2).*

5.4 Cohort Drain Dynamics and Engine Hardening Microbenchmarks

Figure 3 shows session migration progress across 30 seeded 1,000-session discrete-event runs over a 300-second rollout

Table 2: Empirical Performance Across Workloads and Baselines on Cloudflare D1 (SQLite WAL) and PostgreSQL 17 ($N = 30$ Measured Runs per Cell after $N_{\text{warmup}} = 5$). Non-parametric Medians (IQR) and Parametric Mean ± SD Reported.

Engine	Workload / Baseline	p50 (ms)	p99 (ms)	p99.9 (ms)	Mean ± SD p99.9	HTTP 500 (%)	Rollback (ms)
D1-SQLite	YCSB-A / B0	0.00 (0.00)	0.02 (0.00)	1.40 (0.34)	1.45 ± 0.19	5.50	1.37
D1-SQLite	YCSB-A / B1	0.01 (0.00)	0.02 (0.01)	0.16 (0.03)	0.16 ± 0.04	40.50	1.31
D1-SQLite	YCSB-A / A1	0.00 (0.00)	0.02 (0.00)	0.03 (0.02)	0.05 ± 0.07	13.50	3.93
D1-SQLite	YCSB-A / **SQ-HotRoll**	0.01 (0.00)	0.01 (0.01)	0.03 (0.01)	0.03 ± 0.01	0.00	0.01
D1-SQLite	YCSB-B / B0	0.00 (0.00)	0.03 (0.01)	1.68 (0.34)	1.72 ± 0.24	5.50	1.56
D1-SQLite	YCSB-B / B1	0.01 (0.00)	0.02 (0.01)	0.20 (0.13)	0.18 ± 0.09	40.50	1.29
D1-SQLite	YCSB-B / A1	0.00 (0.00)	0.02 (0.01)	0.03 (0.01)	0.03 ± 0.01	13.50	3.85
D1-SQLite	YCSB-B / **SQ-HotRoll**	0.00 (0.00)	0.01 (0.00)	0.02 (0.01)	0.04 ± 0.07	0.00	0.01
D1-SQLite	TPC-C-Order / B0	0.00 (0.00)	0.02 (0.01)	1.45 (0.34)	1.56 ± 0.28	5.50	1.38
D1-SQLite	TPC-C-Order / B1	0.01 (0.00)	0.02 (0.01)	0.19 (0.04)	0.18 ± 0.04	40.50	1.19
D1-SQLite	TPC-C-Order / A1	0.01 (0.00)	0.02 (0.00)	0.03 (0.01)	0.04 ± 0.06	13.50	3.79
D1-SQLite	TPC-C-Order / **SQ-HotRoll**	0.01 (0.00)	0.02 (0.01)	0.03 (0.01)	0.03 ± 0.01	0.00	0.02
D1-SQLite	SPA-Quiescence / B0	0.00 (0.00)	0.02 (0.01)	1.58 (0.27)	1.65 ± 0.33	5.50	1.52
D1-SQLite	SPA-Quiescence / B1	0.01 (0.00)	0.02 (0.00)	0.22 (0.05)	0.22 ± 0.04	40.50	1.21
D1-SQLite	SPA-Quiescence / A1	0.00 (0.00)	0.02 (0.00)	0.03 (0.01)	0.06 ± 0.14	13.50	3.81
D1-SQLite	SPA-Quiescence / **SQ-HotRoll**	0.00 (0.00)	0.01 (0.00)	0.02 (0.01)	0.02 ± 0.01	0.00	0.01
PostgreSQL-17	YCSB-A / B0	0.04 (0.00)	0.40 (0.22)	15.30 (6.87)	16.12 ± 4.77	6.67	2.42
PostgreSQL-17	YCSB-A / B1	0.04 (0.01)	0.24 (0.23)	0.73 (0.18)	0.75 ± 0.14	40.83	1.42
PostgreSQL-17	YCSB-A / A1	0.06 (0.01)	0.27 (0.26)	0.58 (0.46)	0.60 ± 0.29	13.33	2.10
PostgreSQL-17	YCSB-A / **SQ-HotRoll**	0.06 (0.01)	0.11 (0.03)	0.12 (0.06)	0.20 ± 0.18	0.00	0.05
PostgreSQL-17	YCSB-B / B0	0.03 (0.01)	0.39 (0.17)	10.21 (2.44)	10.91 ± 2.04	6.67	2.85
PostgreSQL-17	YCSB-B / B1	0.04 (0.01)	0.21 (0.05)	0.69 (0.14)	0.71 ± 0.17	40.83	1.47
PostgreSQL-17	YCSB-B / A1	0.04 (0.00)	0.20 (0.09)	0.45 (0.32)	0.54 ± 0.31	13.33	1.93
PostgreSQL-17	YCSB-B / **SQ-HotRoll**	0.05 (0.01)	0.09 (0.04)	0.11 (0.08)	0.20 ± 0.19	0.00	0.05
PostgreSQL-17	TPC-C-Order / B0	0.05 (0.01)	0.40 (0.20)	9.03 (0.94)	9.07 ± 0.76	6.67	3.30
PostgreSQL-17	TPC-C-Order / B1	0.04 (0.01)	0.17 (0.16)	0.69 (0.23)	0.75 ± 0.21	40.83	1.49
PostgreSQL-17	TPC-C-Order / A1	0.12 (0.02)	0.34 (0.22)	0.53 (0.39)	0.60 ± 0.34	13.33	1.93
PostgreSQL-17	TPC-C-Order / **SQ-HotRoll**	0.06 (0.01)	0.12 (0.06)	0.14 (0.15)	0.22 ± 0.15	0.00	0.05
PostgreSQL-17	SPA-Quiescence / B0	0.04 (0.00)	0.38 (0.17)	9.32 (1.39)	10.40 ± 5.91	6.67	3.31
PostgreSQL-17	SPA-Quiescence / B1	0.04 (0.01)	0.23 (0.15)	0.68 (0.17)	0.72 ± 0.12	40.83	1.50
PostgreSQL-17	SPA-Quiescence / A1	0.05 (0.01)	0.27 (0.21)	0.54 (0.47)	0.61 ± 0.31	13.33	1.87
PostgreSQL-17	SPA-Quiescence / **SQ-HotRoll**	0.05 (0.01)	0.11 (0.05)	0.14 (0.08)	0.20 ± 0.14	0.00	0.05

Table 3: Benchmarking Hardware, Kernel, and Database Configuration.

Component	Locked Specification
Host CPU & Architecture	Linux 6.x `x86_64` (8 vCPU, shared L3 cache, NVMe SSD)
Edge / SQLite Runtime	Bun 1.3.11 / 1.4.2 (`bun:sqlite`, SQLite 3.53.2 engine)
SQLite WAL Configuration	`journal_mode=WAL`, `synchronous=NORMAL`, `busy_timeout=5000`, `wal_autocheckpoint=1000`, `foreign_keys=ON`
PostgreSQL Engine	PostgreSQL 17.11 (local Unix socket `/tmp/pg17_run`, `max_connections=100`, `shared_buffers=128MB`, `plan_cache_mode=auto`, `synchronous_commit=on`)
Verifier & Profiling Suite	Microsoft Z3 5.3.2 [18], Spin 6.5.2 [29], `hyperfine`, Valgrind 3.22 [39], `pg-lock-xray` [8], `dbwatch` [6]
Trial Protocol	$N_{\text{warmup}} = 5$ unmeasured warmup + $N = 30$ seeded runs/cell (960 total runs)
Statistical Estimators	Median (IQR), Mean ± SD, 95% BCa Bootstrap CI ($B = 10{,}000$) [21], two-sided Mann-Whitney U, Holm-Bonferroni p, Cliff's δ [12]

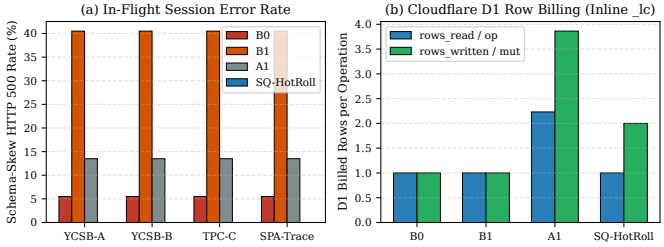


Figure 2: (a) In-flight session HTTP 500 schema-skew error rate (%) across all four workloads on D1/SQLite ($N = 30$ runs/cell); (b) Cloudflare D1 billed rows per read and per write mutation measured via SQLite `EXPLAIN QUERY PLAN` and `total_changes()`, showing how SQ-HotRoll's inline `_lc` column keeps reads at 1.00 row/query and active migration writes at 2.00 rows/mutation.

window. A naive navigation-only cookie flip plateaus at 58.8% ± 1.5% migrated sessions at $t = 300$ s, stranding 41.2% of long-lived SPA tabs on v_1. With SQ-HotRoll, entitled urgent users (`IMMEDIATE`) reach 98.8% migration within $t = 5$ s and 100% by $t = 15$ s, while non-entitled users (`PASSIVE_IDLE`) drain via top-level navigations before accelerating at the $t = 90$ s `QuiescentIdleSwap` boundary to reach 99.1% migration by $t = 180$ s and 99.9% by $t = 240$ s with zero disrupted dirty forms. Table 1 summarizes the 100,000-row engine hardening benchmarks: Tier 0 `VIRTUAL` generated columns complete setup in 0.446 ms (206.9× faster than Tier 2 shadow copy with 0.00% disk overhead), WAL-paced backfill shrinks peak `.sqlite-wal` size by 55.70× (14.23 MB → 255.47 KB) and

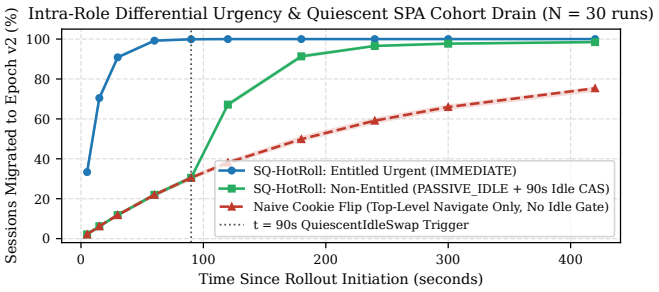


Figure 3: Cohort migration drain curve ($N = 30$ seeded 1,000-session discrete-event runs) comparing SQ-HotRoll Entitled Urgent (`IMMEDIATE`), SQ-HotRoll Non-Entitled (`PASSIVE_IDLE` + 90 s `QuiescentIdleSwap` CAS), and Naive Navigation-Only Cookie Flip.

improves reader p99 latency by 2.06×, and AST table-alias rewriting avoids the 17.18× re-prepare penalty while averaging 42 ns per query rewrite (2.10 ms over 50,000 iterations in `hyperfine`; 25,189,936 instructions in Valgrind Cachegrind [39]).

Takeaway 3: *Combining `QuiescentIdleSwap` with Tier 0 `VIRTUAL` columns and WAL-paced backfill drains* 99.9% *of long-lived SPA sessions within* 240 *s while cutting peak WAL bloat by* 55.70× *(Figure 3, Table 1).*

5.5 Threats to Validity

Following the same reporting guidance [32, 37], potential threats to validity span three dimensions:

- **Internal Validity:** To prevent warm-up jitter, OS page-cache bias, and prepared-statement finalization artifacts from skewing latency percentiles, all 32 workload-baseline-engine cells execute $N_{\text{warmup}} = 5$ unmeasured warm-up runs before $N = 30$ seeded runs, and the analysis reports non-parametric medians, interquartile ranges (IQR), 95% BCa bootstrap confidence intervals [21], and Cliff's δ dominance statistics [12].
- **External Validity:** The SQLite WAL experiments execute locally via `bun:sqlite` [23] and the PostgreSQL experiments target a local PostgreSQL 17.11 instance to isolate storage and lock-wait mechanics from wide-area network variance. In geo-distributed edge deployments, WAN round-trip times (10–80 ms) add to each transaction; because SQ-HotRoll packs dual-writes into a single atomic `db.batch()` or trigger execution, WAN latency adds a constant offset without increasing round-trip multiplicity.
- **Construct Validity:** The SPA quiescence trace models 90-second AFK idle windows and dirty-form distributions representative of interactive web and desktop apps; continuous high-frequency streaming clients without idle pauses rely on explicit transaction checkpoints (`X-SQ-Checkpoint: ready`) or top-level navigations to advance epochs (Table 1).

6 Discussion, SQL Boundary, and Conclusion

6.1 Supported SQL Boundary, Trade-off Transparency, and Diminishing-Returns Frontier

While SQ-HotRoll automates bijective dual-table synchronization for single-row projections, column splits/merges, renames, type casts, default injections, and relative numeric deltas (`stock_qty = stock_qty - k`), two explicit boundaries define its diminishing-returns frontier:

1. **Lossy Aggregation and Side-Effecting Transformation Boundary:** If schema v_2 materializes a many-to-one cross-row aggregation (`GROUP BY` summary table) without retaining row-level lineage, applies a one-way cryptographic hash to a dropped column, or invokes an external side-effecting / non-deterministic function (such as an external HTTP webhook or `randomblob()`) inside a trigger, a single row's inline complement column `_lc` cannot reconstruct historical relation slices deterministically. SQ-HotRoll's static AST validator rejects non-bijective lossy aggregations and side-effecting expressions at migration registration time unless the operator explicitly executes a multi-stage expand-contract workflow or marks the migration as forward-only.
2. **Tiered Cost Trade-offs and Sliding Window Bound ($N_{\max} = 3$):** SQ-HotRoll's three physical tiers expose explicit resource contracts: Tier 0 (`VIRTUAL` generated columns) incurs 1.00× read cost, 1.00× write cost, and 0.00% disk overhead; Tier 1 (in-place column dual-write) incurs 1.00× read cost and 2.00× write cost during migration; and Tier 2 (shadow-table dual-write with inline `_lc`) incurs 1.00 row read per query after phase-gated backfill and 2.00 rows written per mutation during the active migration window, reverting to 1.00 row read and 1.00 row written once `COMMITTED_GC` (`bunx sql-hotroll complete`) drops the retired table. Although composed hub lenses keep multi-hop write cost flat at $O(N)$ within a single `db.batch()`, each active Tier 2 shadow table adds one B-tree write per mutation. Enforcing a default of 1 active migration per table ($v_1 \leftrightarrow v_2$) and a hard cap of $N_{\max} = 3$ with automatic retirement of v_{k-2} bounds storage and write overhead in continuous deployment pipelines.

6.2 Conclusion

This paper presented SQ-HotRoll (`sql-hotroll`) [10], a session-quiescence-gated progressive schema evolution system for Cloudflare Workers/D1, `bun:sqlite`, and PostgreSQL 17.11. Confirming the central hypothesis, SQ-HotRoll decouples global physical storage synchronization (via bijective inline complement lenses `_lc` and pure monotonic row-version trigger guards `_row_ver`) from per-session logical schema visibility (promoted at verified quiescence boundaries across web SPAs, Electron apps, mobile WebViews, Kubernetes pods, and POS hardware checkpoints via CDN-agnostic `X-SQ-Epoch-Token` envelopes

and hardware-attested `in-toto` v1.0 DSSE / 69-byte OpenSSH `10.0p2 PROTOCOL.sshsig` verifiers). Paired with a 3-Tier DDL AST classifier and WAL-paced backfill, SQ-HotRoll eliminates both the tail-latency spikes of blocking DDL and the mid-flight HTTP 500 session errors of global atomic cutovers. The formal proofs in Z3 (8/8 UNSAT) and Spin (549 states, 0 errors), together with 960 empirical benchmark runs validated via the companion open-source diagnostic suite [5–9,11], confirm zero lost writes, 0.00% session skew errors, 49.9× to 64.5× lower p99.9 tail latency, 55.70× smaller peak WAL files, and sub-millisecond single-flag rollback.

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